

CIS41032: Practical for Advance Database Management System
Department of ICT
Faculty of Technology
South Eastern University of Sri Lanka

Academic Year: 2020/2021

Lab Sheet 08

Title: Work with Aggregation Pipeline.

Objective:

- Understand and Apply MongoDB Aggregation Framework.
- Differentiate Between Key Aggregation Stages.
- Work with Accumulator Operators

Exercise 1:

Database: Garment

OrderID	CustomerName	Product	Category	Quantity	Price	OrderDate
G001	Anu	T-Shirt	Tops	3	20	2023-01-10
G002	Ravi	Jeans	Bottoms	1	40	2023-01-15
G003	Sara	Jacket	Outerwear	2	60	2023-01-20
G004	Anu	Skirt	Bottoms	1	25	2023-01-25
G005	Nina	Hoodie	Outerwear	1	55	2023-02-01
G006	Ravi	Shorts	Bottoms	2	30	2023-02-10
G007	Sara	Blouse	Tops	2	35	2023-02-14
G008	Nina	Coat	Outerwear	1	90	2023-02-20
G009	Anu	Shirt	Tops	2	25	2023-03-01
G010	Ravi	Skirt	Bottoms	1	25	2023-03-03
G011	Sara	Jacket	Outerwear	1	60	2023-03-05
G012	Nina	T-Shirt	Tops	2	20	2023-03-10

1. Insert all data in to the collection “garmentOrders”

2. Show all orders in the "Tops" category.
3. Calculate total revenue from each category.
4. Find the highest quantity of any product ordered by a customer.
5. Calculate the total amount spent by each customer.
6. Show the average price of products in each category.
7. Group by category and count how many products were sold.
8. Sort customers by total amount spent in descending order.
9. Find all orders placed after February 1, 2023.
10. Group products by category and find the maximum price.
11. Save all "Outerwear" orders to a new collection outerwearOrders.

Exercise 02:

Database: Factory

SupplyID	SupplierName	Material	MaterialType	UnitsSupplied	UnitPrice	SupplyDate
S001	TexWorld	Cotton Roll	Fabric	100	5	2023-01-05
S002	SpinFast	Zippers	Accessories	200	0.5	2023-01-10
S003	TexWorld	Buttons	Accessories	500	0.2	2023-01-15
S004	ThreadX	Thread Pack	Thread	300	1	2023-01-20
S005	Colorify	Dye Pack	Dye	50	8	2023-01-25
S006	TexWorld	Cotton Roll	Fabric	150	5	2023-02-01
S007	SpinFast	Zippers	Accessories	300	0.5	2023-02-05
S008	ThreadX	Thread Pack	Thread	400	1	2023-02-15
S009	TexWorld	Linen Roll	Fabric	100	6	2023-02-20
S010	Colorify	Dye Pack	Dye	60	8	2023-03-01
S011	ThreadX	Thread Pack	Thread	350	1	2023-03-05
S012	SpinFast	Buttons	Accessories	400	0.2	2023-03-10

1. Insert all data in to the collection “**factorySupplies**”
2. Show all supplies in the "Accessories" material type.
3. Calculate total cost (Units $\times$ Price) of supplies per material.
4. Group by SupplierName and find the total number of units supplied.
5. Find the average unit price for each material type.
6. List the top 2 materials with the highest total cost.
7. Count the number of different materials supplied by each supplier.
8. Show all supply records between Jan 20 and Feb 20, 2023.
9. Group by MaterialType and find the max and min unit prices.
10. Calculate the total cost per supplier and sort by cost descending.
11. Output all “Fabric” supplies into a new collection fabricSupplies.